



JAIN UNIVERSITY

Declared as Deemed-to-be University u/s 3 of the UGC Act, 1956

Department of Computer Science and Information Technology

SEMISTER: I

SPECILISATION: DATA ANALYTICS

JUUG23BCA528059

Lab Experiments- 1&3

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Submitted to

Ms. Kriti Kausik

INDEX

Sl No.	Content	Page No.
1.	Resume	3
2.	Excel: Calculate sum, max, min, avg, std, range, cv for B1:B8	4-5
3.	Create worksheet with roll numbers, random marks, average marks, pass/fail status, and grades	6-8
4.	Worksheet	9-10
5.	Row	10
6.	Column	11
7.	Terms and Functions	12

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Education

JAIN (DEEMED-TO-BE UNIVERSITY)

Bengaluru, IND

B.C.A Degree Candidate in Artificial Intelligence

Expected March 2026

Relevant Coursework: Pretrained Language Models - Analyze models like BERT, GPT-2/3, T5 etc. and approaches for advanced transfer learning.

ALVAS CENTRAL SCHOOL

Mangalore, IND

Major: Mathematics, GPA: 3.3

March 2021

Class Representative - Elected by peers; led meetings and communicated student needs. Head of Club House Red - Organized events and led a team of 170 students. Excellence in Abacus Award (2020) - Top score in school-wide competition. Best Creative Project, Science Forum (2020) - Honored for innovative science fair project.

Experience

HUGGING FACE

New York, U.S

Research Intern

April 2023

- Fine-tuned LLMs including LLa and BERT to optimize model accuracy.
- Employed techniques like conditional training and knowledge distillation.
- Collaborated with researchers to test new techniques and analyze model capabilities.

AVERSA AI

Tel Aviv, IL

Research Intern

November 2022

- Spearheaded innovative prompting techniques and jailbreaking strategies to optimize large language models (LLMs).
- Conducted rigorous experimentation and data analysis to refine LLM behavior and capabilities.

Skills & Extracurricular Activities

Skills: Experience training and fine-tuning transformer models like BERT, GPT-2/3, T5, DALL-E. Knowledge of NLP techniques such as tokenization, embeddings, attention mechanisms. Python, TensorFlow, PyTorch, Hugging Face Transformers, etc.

Extracurricular Activities: Reading Non-Fiction, Browsing Internet, listening to podcasts, playing chess, Analysis of yearly reports.

3.1 How to calculate the following values in Excel using formulas for a given range of values from B1 to B8:

- 1) The sum of the values in cells B1 to B8, in cell B10.**
- 2) The maximum value in cells B1 to B8, in cell B11.**
- 3) The minimum value in cells B1 to B8, in cell B12.**
- 4) The average of the values in cells B1 to B8, in cell B13.**
- 5) The standard deviation of the values in cells B1 to B8, in cell B14.**
- 6) The sum of the maximum and minimum values in cells B1 to B8, in cell B15.**
- 7) The result of dividing the standard deviation (from cell B14) by the sum of the maximum and minimum values (from cell B15), in cell B16.**

1. Open a new Excel workbook
2. Insert a new worksheet and rename it to "Sheet1."
3. In cell B1, type the value 1000
4. Click on cell B1 to select it
5. Move your mouse to the bottom right corner of cell B1 until you see a + sign
6. Click and drag down to cell B8 to autofill the range B1:B8 with 1000
7. In cell A10, type "SUM"
8. In cell B10, enter the formula =SUM(B1:B8)
9. In cell A11, type "MAXIMUM"
10. In cell B11, enter the formula =MAX(B1:B8)
11. In cell A12, type "MINIMUM"
12. In cell B12, enter the formula =MIN(B1:B8)
13. In cell A13, type "AVERAGE"
14. In cell B13, enter the formula =AVERAGE(B1:B8)
15. In cell A14, type "STANDARD DEVIATION"
16. In cell B14, enter the formula =STDEV.S(B1:B8)
17. In cell A15, type "SUM OF MAXIMUM AND MINIMUM"
18. In cell B15, enter the formula =SUM(B11+B12)
19. In cell A16, type "DIVIDE"
20. In cell B16, enter the formula =B14/B15

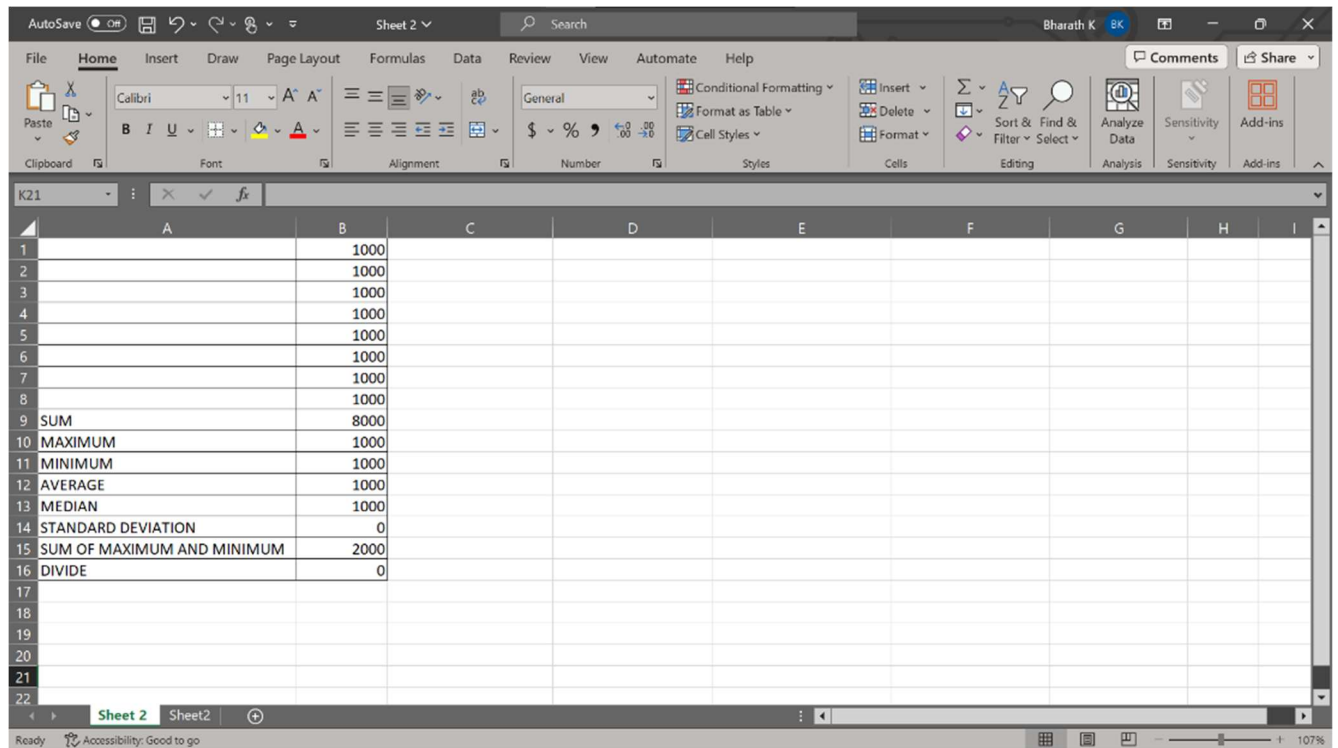


Image 1- (Calculating the sum, max, min, avg, std, sum of min and max, division of std to sum of min and max)

3.2 How to create a worksheet with roll numbers, random marks for Subject 1 and Subject 2, average marks, pass/fail status, and grades for all students in Excel?

Step 1: Enter Headings

1. In cell A1, enter "Roll No."
2. In cell B1, enter "Subject 1."
3. In cell C1, enter "Subject 2."
4. In cell D1, enter "Average."
5. In cell E1, enter "Result."
6. In cell F1, enter "Grade."

Step 2: AutoFill Roll Numbers

1. In cell A2, enter the first roll number 1.
2. In cell A3, enter the next roll number 2.
3. Select cells A2 and A3.
4. Hover over the bottom-right corner of the selected cells (you'll see a small square called the "Fill Handle").
5. Click and drag the Fill Handle down to autofill the roll numbers for all 50 students.

Step 3: Enter Random Marks for Subject 1 and Subject 2 with Formulas

1. In cell B2, enter the formula to generate random marks for Subject 1:

=RANDBETWEEN(0, 100)

This formula generates a random number between 0 and 100. You can copy and paste this formula into cell B3 and drag it down to autofill random marks for all students in column B.

2. Similarly, in cell C2, enter the formula to generate random marks for Subject 2:

=RANDBETWEEN(0, 100)

Copy and paste this formula into cell C3 and drag it down to autofill random marks for all students in column C.

Step 4: Calculate Average Marks

1. In cell D2, enter the formula to calculate the average of the marks for Subject 1 and Subject 2 for the first student:

=(B2 + C2) / 2 or =AVERAGE(B2, C2)

This formula calculates the average of the marks in cells B2 and C2. Copy and paste this formula into cell D3 and drag it down to calculate the average marks for all students in column D.

Step 5: Determine Pass or Fail Status

1. In cell E2, enter the formula to determine if the student passed or failed:

=IF(D2 >= 40, "PASS", "FAIL")

This formula checks if the average marks in cell D2 are greater than or equal to 40. If yes, it returns "PASS," otherwise, it returns "FAIL." Copy and paste this formula into cell E3 and drag it down to apply it to all students in column E.

Step 6: Calculate Grades for Passed Students

1. In cell F2, determine the grade for passed students based on the average marks:

=IF(E2="FAIL","IV",IF(D2>=60,"I",IF(D2>=50,"II",IF(D2>=40,"III"))))

This nested IF formula checks if the student passed (cell E2). If yes, it calculates the grade based on the average marks in cell D2. If the student failed, it assigns the grade "IV." Copy and paste this formula into cell F3 and drag it down to calculate grades for all students in column F.

Now, you have a worksheet with roll numbers, random marks for Subject 1 and Subject 2, average marks, pass/fail status, and grades for all students.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
	Roll No.	Subject 1	Subject 2	Average	Result	Grade																
2	1	99	36	67.5	PASS	I																
3	2	33	72	52.5	PASS	II																
4	3	94	64	79	PASS	I																
5	4	70	95	82.5	PASS	I																
6	5	56	35	45.5	PASS	III																
7	6	97	99	98	PASS	I																
8	7	20	59	39.5	FAIL	IV																
9	8	44	53	48.5	PASS	III																
10	9	74	47	60.5	PASS	I																
11	10	87	59	73	PASS	I																
12	11	49	46	47.5	PASS	III																
13	12	37	66	51.5	PASS	II																
14	13	54	72	63	PASS	I																
15	14	65	69	67	PASS	I																
16	15	80	88	84	PASS	I																
17	16	44	70	57	PASS	II																
18	17	48	87	67.5	PASS	I																
19	18	98	44	71	PASS	I																
20	19	34	89	61.5	PASS	I																
21	20	86	94	90	PASS	I																
22	21	80	54	67	PASS	I																
23	22	48	96	72	PASS	I																
24	23	48	48	48	PASS	III																
25	24	91	52	71.5	PASS	I																
26	25	59	96	77.5	PASS	I																
27	26	22	76	49	PASS	III																

Image 2 - (Creating worksheet with roll numbers, random marks to subject 1 & subject 2, average marks, pass/fail status, and grades; from roll number 1 to 26)

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
27	26	22	76	49	PASS	III																
28	27	57	56	56.5	PASS	II																
29	28	78	49	63.5	PASS	I																
30	29	84	96	90	PASS	I																
31	30	63	96	79.5	PASS	I																
32	31	25	51	38	FAIL	IV																
33	32	85	75	80	PASS	I																
34	33	85	36	60.5	PASS	I																
35	34	55	97	76	PASS	I																
36	35	86	81	83.5	PASS	I																
37	36	76	35	55.5	PASS	II																
38	37	91	63	77	PASS	I																
39	38	30	86	58	PASS	II																
40	39	54	48	51	PASS	II																
41	40	64	70	67	PASS	I																
42	41	95	98	96.5	PASS	I																
43	42	45	95	70	PASS	I																
44	43	74	41	57.5	PASS	II																
45	44	69	88	78.5	PASS	I																
46	45	49	49	49	PASS	III																
47	46	22	66	44	PASS	III																
48	47	61	63	62	PASS	I																
49	48	99	43	71	PASS	I																
50	49	45	60	52.5	PASS	II																
51	50	67	86	76.5	PASS	I																
52																						
53																						

Image 3 - (Creating worksheet with roll numbers, random marks to subject 1 & subject 2, average marks, pass/fail status, and grades; from roll number 26 to 50)

3.3 (a)- WORKSHEET

Define worksheet with diagram.

A worksheet in Excel is a single grid-like page within a workbook where data is organized in rows and columns for tasks like data entry, calculations, and analysis.

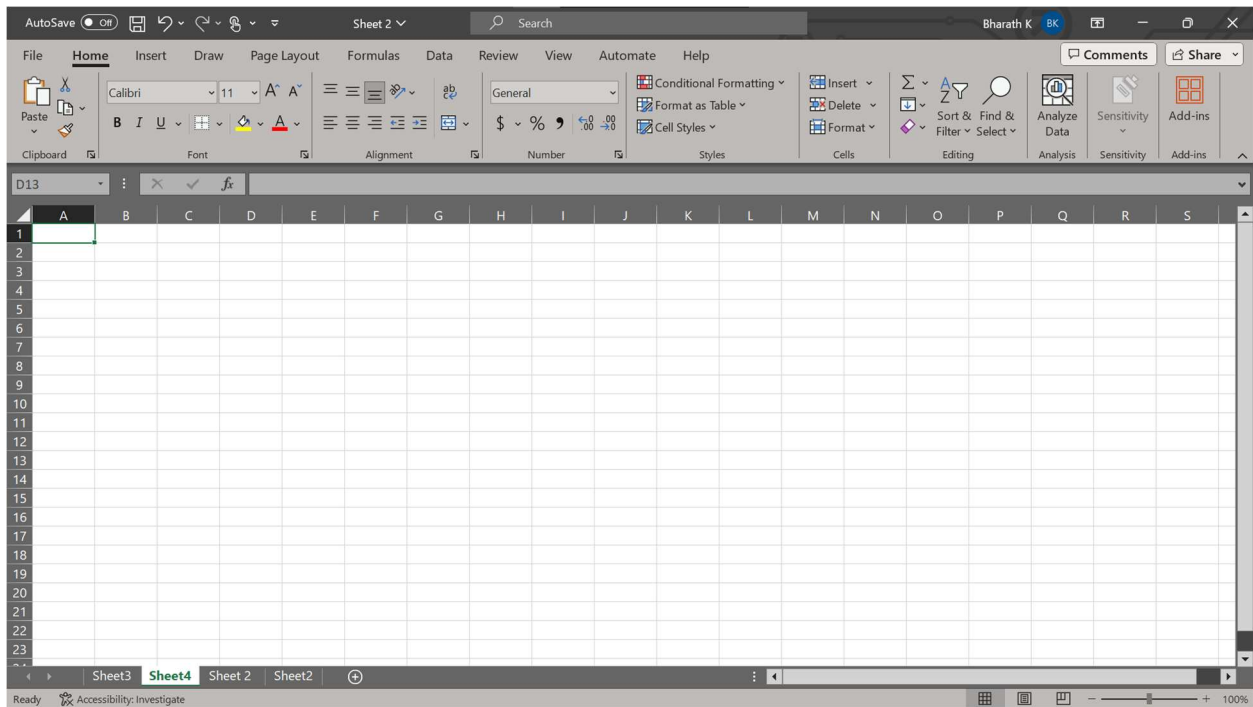


Image 5 - (Example of an empty excel worksheet)

How to select worksheet?

To select a worksheet in Excel, follow these steps:

1. Open your Excel workbook, which may contain multiple worksheets.
2. Look at the bottom of the Excel window, where you'll see tabs labeled with the names of each worksheet in the workbook. These tabs are typically located in a sheet navigation bar.
3. Click on the tab of the worksheet you want to select. The selected worksheet will become the active one, and its contents will be displayed in the Excel window.

How to insert worksheet.

To insert a new worksheet in Excel, follow these steps:

1. Open your Excel workbook.

2. Look at the bottom of the Excel window, where you'll see tabs labeled with the names of each existing worksheet in the workbook. These tabs are typically located in a sheet navigation bar.

3. To insert a new worksheet, right-click on an existing worksheet tab. This will open a context menu.

4. From the context menu, select "Insert."

5. A dialog box will appear, allowing you to specify where you want to place the new worksheet. You can choose to insert it before the selected worksheet or after the selected worksheet. You can also rename the new worksheet if desired.

6. Click the "OK" button.

A new worksheet will be inserted into your Excel workbook, and you can begin working with it immediately.

The new worksheet will be named with a default name like "Sheet1," "Sheet2," and so on, but you can rename it by right-clicking on its tab and selecting "Rename" from the context menu.

Rename a worksheet.

To rename a worksheet in Excel, follow these steps:

1. Open your Excel workbook.

2. Look at the bottom of the Excel window, where you'll see tabs labeled with the names of each worksheet in the workbook. These tabs are typically located in a sheet navigation bar.

3. Right-click on the worksheet tab that you want to rename. This will open a context menu.

4. From the context menu, select "Rename." Alternatively, you can also double-click on the worksheet tab to activate the renaming mode.

5. The worksheet tab label will become editable, allowing you to enter a new name for the worksheet.

6. Type in the desired name for the worksheet and press "Enter" on your keyboard to save the new name.

The worksheet will now be renamed with the name you provided, and you can easily identify and work with it using the new name.

3.3 (b)- ROW

Define row, no. of row in worksheet with diagram.

A row in an Excel worksheet is a horizontal line of cells that is identified by a numerical label. Rows are used to organize and display data horizontally in a spreadsheet. Each row is numbered sequentially, starting with "1" for the first row. In the latest versions of Excel (Excel 2007 and later, including Excel 2019 and Microsoft 365), a worksheet can have a maximum of 1,048,576 rows.

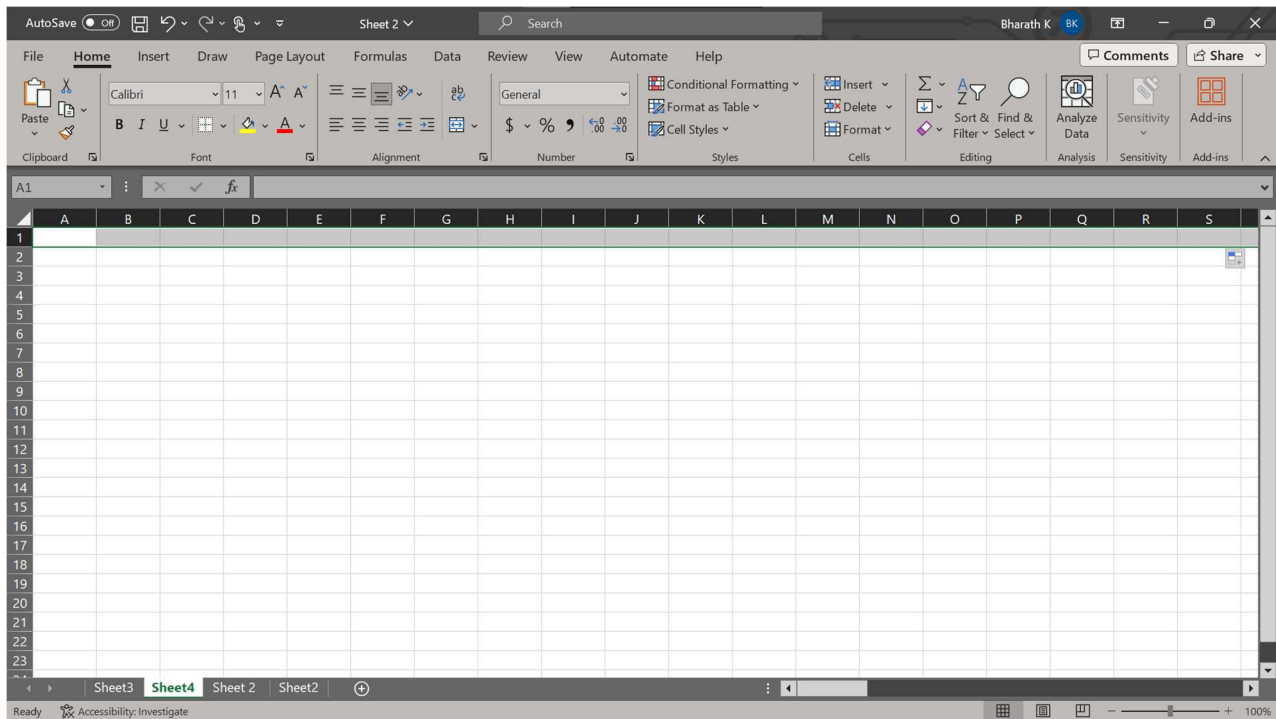


Image 6 - (Selected blocks are the examples of Row's in a worksheet)

3.3 (c)-COLUMN

Define column, no. of column in worksheet with diagram.

A column in an Excel worksheet is a vertical arrangement of cells that is identified by a lettered label. Columns are used to organize and display data vertically in a spreadsheet. Each column is labeled with a letter, starting with "A" for the first column, "B" for the second column, and so on.

The number of columns in an Excel worksheet is consistent across all versions of Excel, including the latest versions like Excel 2019 and Microsoft 365. In every Excel worksheet, you have a total of 16,384 columns, which are labeled from "A" to "XFD".

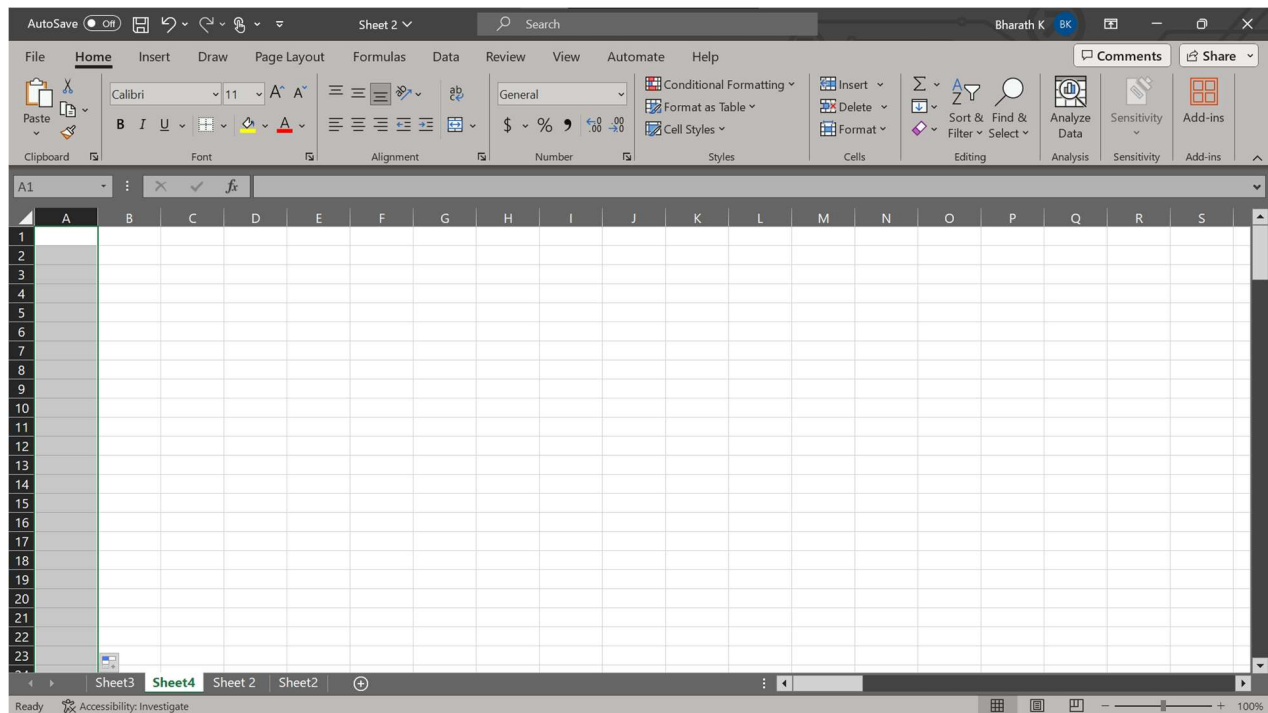


Image 7 - (Selected blocks are the examples of Column's in a worksheet)

3.3 (d)-Create one worksheet in excel and identify all the functions and terms.

1. Worksheet: This is the tab at the bottom of the Excel window where you can enter and manipulate data. In this example, let's call it "Sample Worksheet."
2. Cell: A cell is the intersection of a row and a column in the worksheet where data can be entered and displayed. Cells are identified by a unique address consisting of a column letter and row number. For example, cell A1 is in the first column and first row.
3. Column Header: The letters at the top of each column represent column headers. They help you identify and reference specific columns. For example, "A" is the column header for the first column.
4. Row Header: The numbers on the left side of each row are row headers. They help you identify and reference specific rows. For example, "1" is the row header for the first row.
5. Data: These are the values you enter into the cells. You can input numbers, text, dates, or formulas into cells to store and manipulate data.
6. Formula Bar: The formula bar is located just below the toolbar and displays the contents of the active cell. You can enter or edit data and formulas in this bar.
7. Formula: A formula is an equation that performs calculations on data in one or more cells. Formulas in Excel start with an equal sign (=). For example, `=A1+B1` is a formula that adds the values in cells A1 and B1.
8. Function: Functions are pre-built formulas in Excel that perform specific calculations or tasks. They start with the equal sign followed by the function name and arguments enclosed in parentheses. For example, `=SUM(A1:A5)` is a function that calculates the sum of values in cells A1 to A5.
9. AutoFill Handle: The small square at the bottom-right corner of the active cell is called the AutoFill handle. You can use it to copy or fill values, formulas, or patterns into adjacent cells.
10. Workbook: A workbook is the entire Excel file containing multiple worksheets. In this example, you have one worksheet within a workbook.
11. Sheet Tab: These are tabs at the bottom of the workbook window that allow you to navigate between different worksheets within the same workbook.
12. Name Box: The name box displays the address or name of the active cell. You can use it to quickly navigate to specific cells.

This are some of the fundamental terms and functions in Excel.